

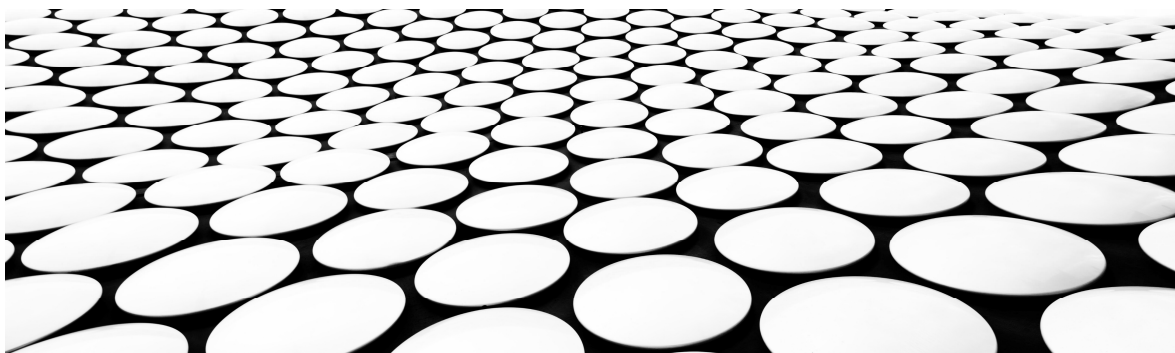


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TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN
KHOA ĐIỆN TỬ - VIỄN THÔNG

THIẾT KẾ VI MẠCH ĐIỆN TỬ

CHAPTER 1: CMOS TECHNOLOGY AND FABRICATION

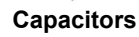
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CONTENT

- Introduction to Semiconductor
- Introduction to MOSFET
- CMOS Technology
- CMOS Fabrication Process

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3

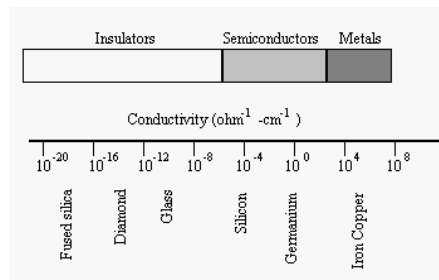
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- Silic (Si) Elements

4

WHAT IS A SEMICONDUCTOR

Range of Conduction



- The semiconductors fall somewhere midway between conductors and insulators.
- Semiconductors have special electronic properties which allow them to be insulating or conducting depending on their composition.

5

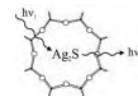
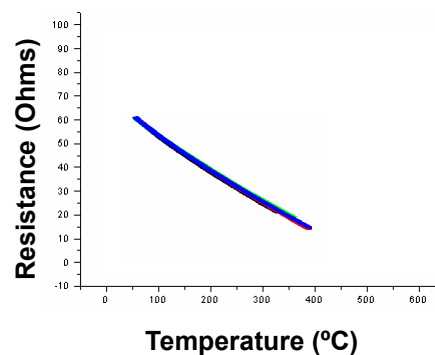
WHAT IS A SEMICONDUCTOR

- 1833, Faraday discovers that electrical resistivity decreases as temperature increases in Silver Sulfide (Ag₂S). This is the first investigation of a semiconductor.

1833



Michael Faraday



6

WHAT IS A SEMICONDUCTOR

- Experiments on Metals vs. Semiconductors

<i>Temperature</i>	<i>Copper</i>	<i>Germanium</i>
<i>0°C</i>	<i>31Ω</i>	<i>5.2Ω</i>
<i>25°C</i>	<i>33Ω</i>	<i>4.2Ω</i>
<i>50°C</i>	<i>37Ω</i>	<i>1.2Ω</i>
<i>75°C</i>	<i>41Ω</i>	<i>0.63Ω</i>
<i>100°C</i>	<i>44Ω</i>	<i>.029Ω</i>

7

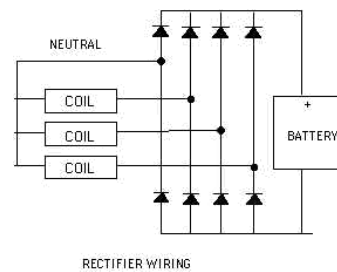
WHAT IS A SEMICONDUCTOR

1874



Ferdinand Braun

- The first semiconductor device was born.
- Radio receivers required a device called a rectifier to detect signals.
- He used the rectifying properties of the galena crystal, a semiconductor material composed of lead sulfide, to create the **cat's whisker diode** for this purpose.



8

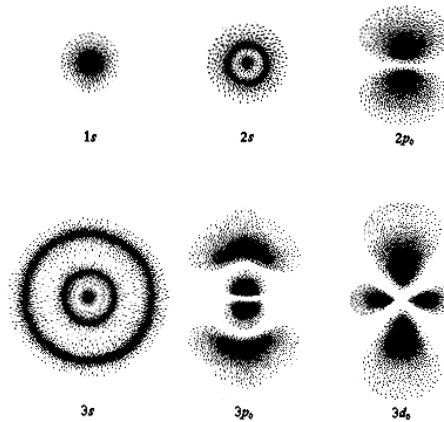
WHAT IS A SEMICONDUCTOR

- Applied quantum mechanics to solids, helping explain the conduction of electricity in semiconductors.

1927



Sommerfeld



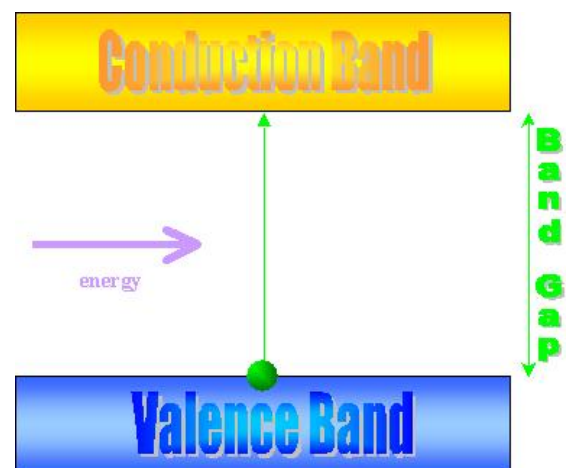
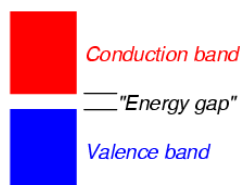
Bloch

9

WHAT IS A SEMICONDUCTOR

Scientific Principle of Conduction

- The highest occupied energy band is called the **valence band**. Most electrons remain bound to the atoms in this band.
- The **conduction band** is the band of orbitals that are high in energy and are generally empty. It is the band that accepts the electrons from the valence band.
- The "leap" or "Energy" required for electrons from the Valence Band to enter the Conduction Band.



10

WHAT IS A SEMICONDUCTOR

■ Examples

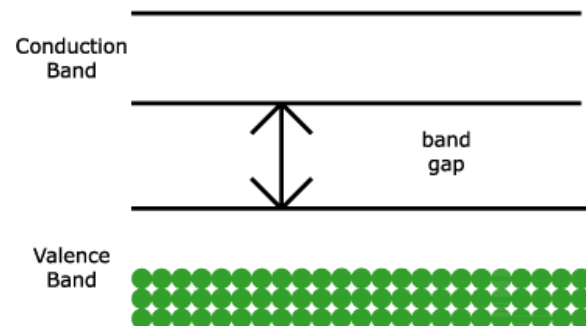
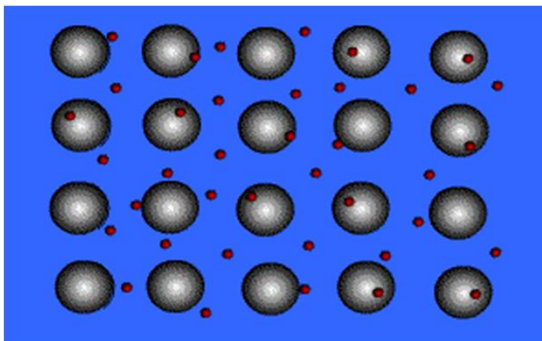


11

WHAT IS A SEMICONDUCTOR

Conductors

- In a conductor, electrons can move freely among these orbitals within an energy band as long as the orbitals are not completely occupied.
- In conductors, the valence band is not full.



12

WHAT IS A SEMICONDUCTOR

Conductors



13

WHAT IS A SEMICONDUCTOR

Conductors

- Also in conductors, the energy gap is nonexistent or relatively small.

*Overlap permits
electrons to freely
drift between bands*



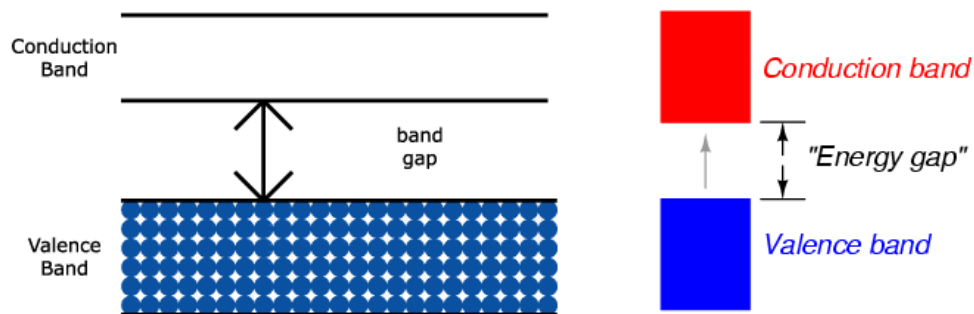
Multitudes of atoms
in close proximity

14

WHAT IS A SEMICONDUCTOR

Insulator

- In insulators, the valence band is full.
- Also in insulators, the energy gap is relatively large.



15

WHAT IS A SEMICONDUCTOR

Insulator

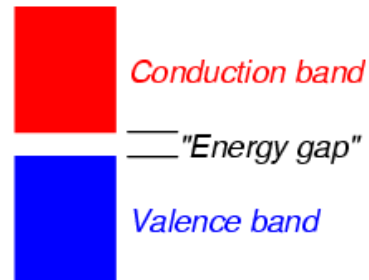
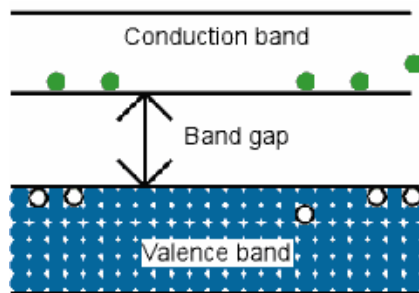


16

WHAT IS A SEMICONDUCTOR

Semiconductor

- In semiconductors, the valence band is full but the energy gap is intermediate.
- Only a small leap is required for an electron to enter the Conduction Band.



17

WHAT IS A SEMICONDUCTOR

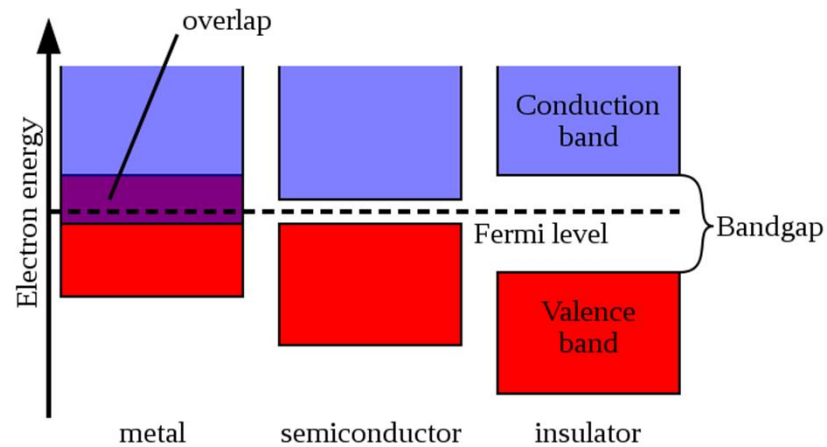
Semiconductor



18

WHAT IS A SEMICONDUCTOR

■ Band Diagrams

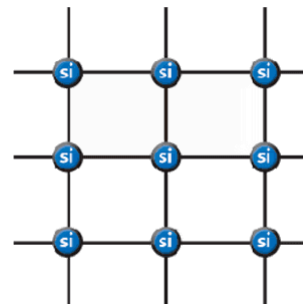


19

WHAT IS A SEMICONDUCTOR

- Silicon is one of chemical elements in Semiconductor

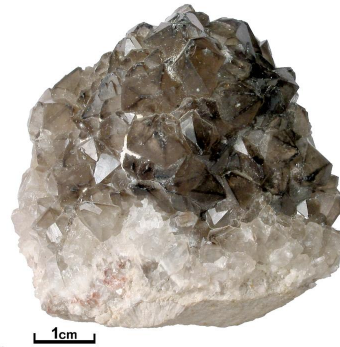
5 B Boron 2.34	6 C Carbon 2.62	7 N Nitrogen 1.251
13 Al Aluminum 2.70	14 Si Silicon 2.33	15 P Phosphorus 1.82
31 Ga Gallium 5.91	32 Ge Germanium 5.32	33 As Arsenic 5.72



20

WHAT IS A SEMICONDUCTOR

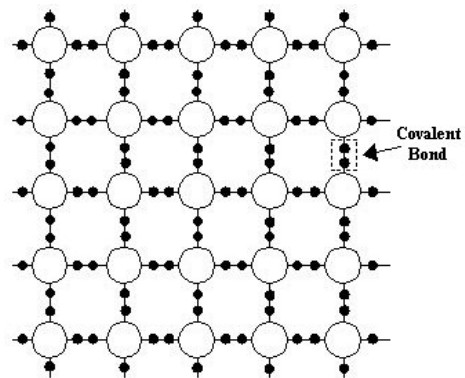
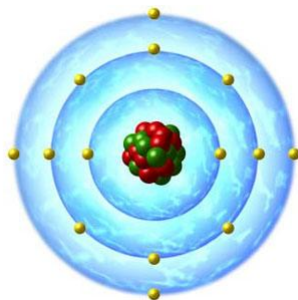
- Silicon is a very common element, the main element in sand & quartz.



21

WHAT IS A SEMICONDUCTOR

- Silicon Arrangement

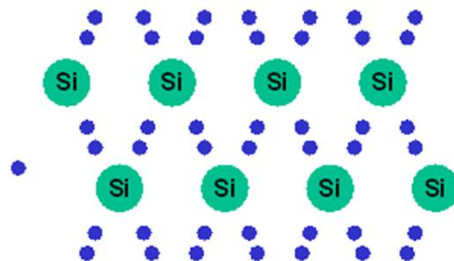
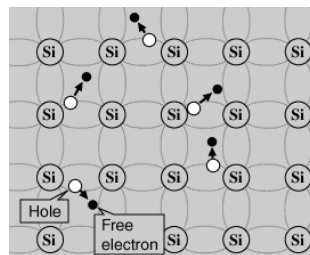
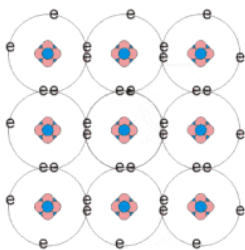


22

WHAT IS A SEMICONDUCTOR

■ Intrinsic Silicon

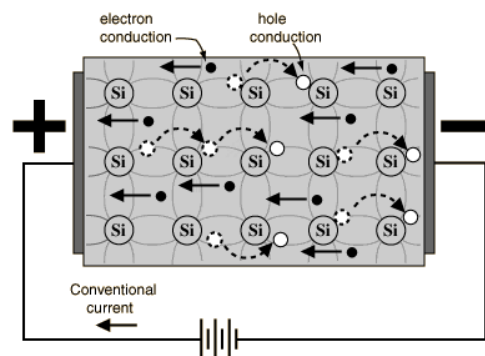
- A silicon crystal is different from an insulator.
- At any temperature above absolute zero temperature, there is a finite probability that an electron in the lattice will be knocked loose from its position.
- The electron in the lattice knocked loose from its position leaves behind an electron deficiency called a "hole".



WHAT IS A SEMICONDUCTOR

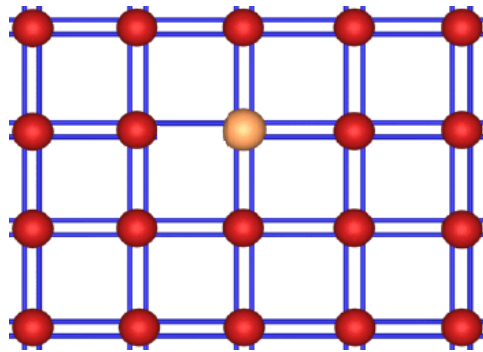
■ Current Flow

- If a voltage is applied, then both the electron and the hole can contribute to a small current flow.



WHAT IS A SEMICONDUCTOR

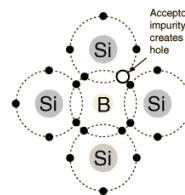
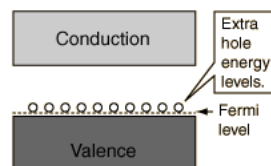
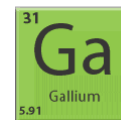
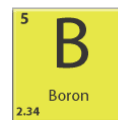
- **Doping**
- Doping (adding an impurity) can produce 2 types of semi-conductors depending upon the element added.
- There are p-type and n-type dopings



25

WHAT IS A SEMICONDUCTOR

- **P-type Doping**
- In P-type doping, boron or gallium is the dopant.
- Boron and gallium each have only three outer electrons.
- When mixed into the silicon lattice, they form "holes" in the lattice where a silicon electron has nothing to bond to.
- The absence of an electron creates the effect of a positive charge, hence the name P-type.
- Holes can conduct current. A hole happily accepts an electron from a neighbor, moving the hole over a space. P-type silicon is a good conductor.

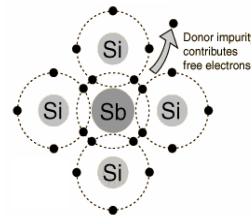
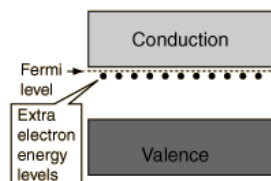
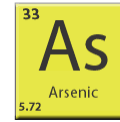
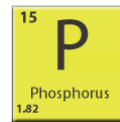


26

WHAT IS A SEMICONDUCTOR

■ N-type Doping

- In N-type doping, phosphorus or arsenic is added to the silicon in small quantities
- Phosphorus and arsenic each have five outer electrons, so they're out of place when they get into the silicon lattice.
- The fifth electron has nothing to bond to, so it's free to move around.
- It takes only a very small quantity of the impurity to create enough free electrons to allow an electric current to flow through the silicon. N-type silicon is a good conductor.
- Electrons have a negative charge, hence the name N-type.

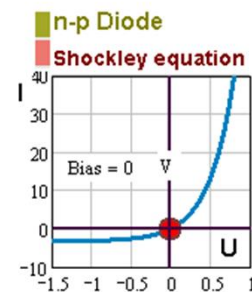
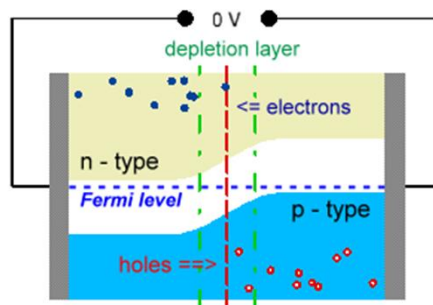


27

WHAT IS A SEMICONDUCTOR

■ P-N Junction

- We create a p-n junction by joining together two pieces of semiconductor, one doped n-type, the other p-type.

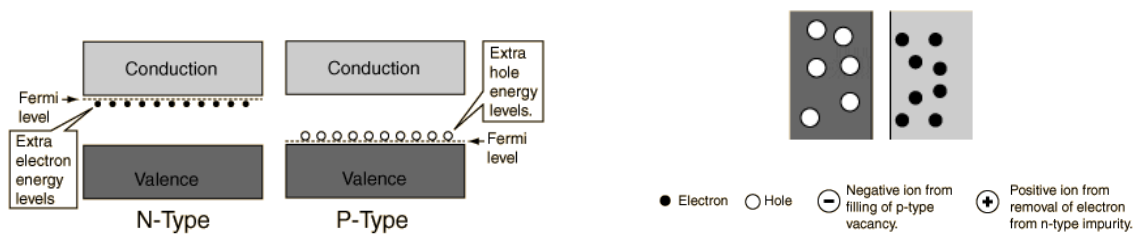


28

WHAT IS A SEMICONDUCTOR

P-N Junction

- In the n-type region there are extra electrons and in the p-type region, there are holes from the acceptor impurities.
- In the p-type region there are holes from the acceptor impurities and in the n-type region there are extra electrons.

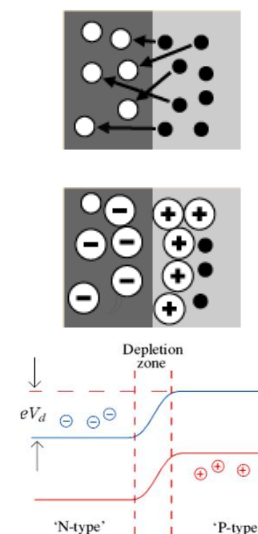


29

WHAT IS A SEMICONDUCTOR

P-N Junction

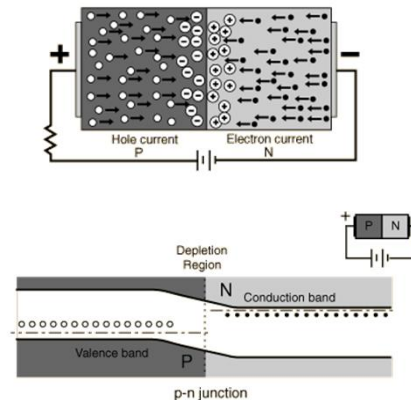
- When a p-n junction is formed, some of the electrons from the n-region which have reached the conduction band are free to diffuse across the junction and combine with holes.
- Filling a hole makes a negative ion and leaves behind a positive ion on the n-side. A space charge builds up, creating a depletion region.
- A depletion zone forms around the junction (the join) between the two materials. This zone controls the behavior of the diode.



30

WHAT IS A SEMICONDUCTOR

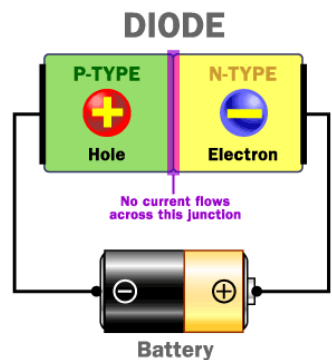
- P-N Junction (Forward Biasing)
- Forward biasing the p-n junction drives holes to the junction from the p-type material and electrons to the junction from the n-type material.
- At the junction the electrons and holes combine so that a continuous current can be maintained.



31

WHAT IS A SEMICONDUCTOR

- P-N Junction (Diode)
- A diode is the simplest possible semiconductor device.



32

WHAT IS A SEMICONDUCTOR

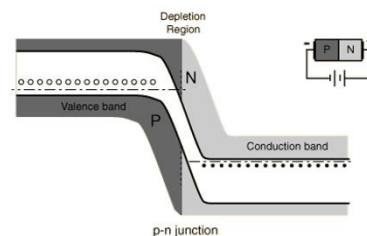
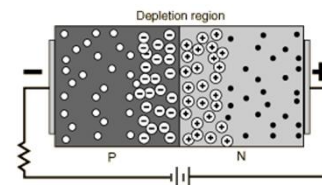
- P-N Junction (Diode)
- A diode allows current to flow in one direction but not the other.
- If you apply enough reverse voltage, the junction breaks down and lets current through.



33

WHAT IS A SEMICONDUCTOR

- P-N Junction (Reverse Biasing)
- The application of a reverse voltage to the p-n junction will cause a transient current to flow as both electrons and holes are pulled away from the junction.
- When the potential formed by the widened depletion layer equals the applied voltage, the current will cease except for the small thermal current.

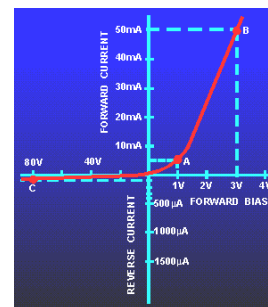
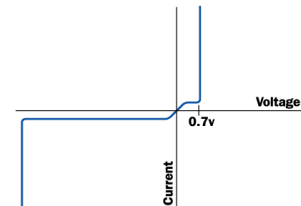


34

WHAT IS A SEMICONDUCTOR

■ P-N Junction (Diode Characteristic)

- When forward-biased, there is a small amount of voltage necessary to get the diode going. In silicon, this voltage is about 0.7 volts → This voltage is needed to start the hole-electron combination process at the junction.
- When reverse-biased, an ideal diode would block all current. A real diode lets perhaps 10 microamps through -- not a lot, but still not perfect.



35

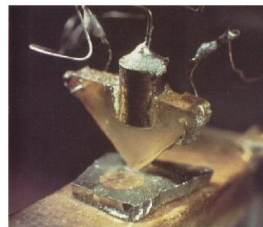
WHAT IS A SEMICONDUCTOR

■ Transistor

- 1947, Bardeen, Shockley and Brattain invented a transistor at Bell Labs. they were trying to understand the nature of the electrons at the interface between a metal and a semiconductor (germanium).
- The 1st transistor consisted of a plastic triangle lightly suspended above a germanium crystal which itself was sitting on a metal plate attached to a voltage source. A strip of gold was wrapped around the point of the triangle with a tiny gap cut into the gold at the precise point it came in contact with the germanium crystal.

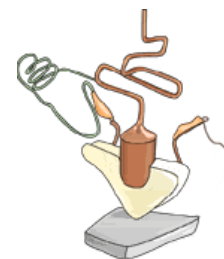


John Bardeen, William Shockley and Walter Brattain



The 1st Transistor

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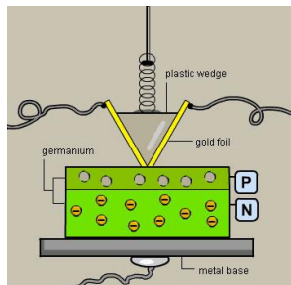


36

WHAT IS A SEMICONDUCTOR

■ Transistor

- The Germanium acted as a semiconductor so that a small electric current entering on one side of the gold strip came out the other side as a proportionately amplified current.

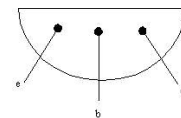
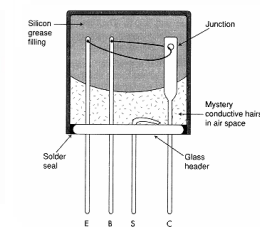
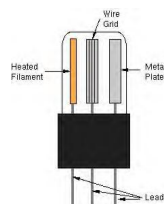


37

WHAT IS A SEMICONDUCTOR

■ Transistor

- Transistors didn't need time to "warm up" like the heaters in vacuum tube circuits
- The transistor is a three-terminal device (B, C, E or D, G, S) and consists of three distinct layers.
- A transistor is a semiconductor device commonly used to amplify or switch electronic signals.

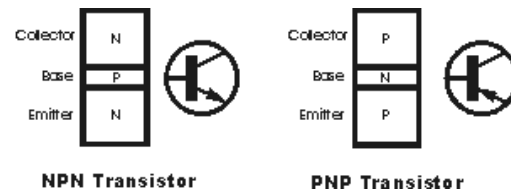


38

WHAT IS A SEMICONDUCTOR

■ Transistor

- Two of them are doped to give one type of semiconductor and the there is the opposite type, i.e. two may be n-type and one p-type, or two may be p-type and one may be n-type.
- For BJT, they are designated either P-N-P (PNP) types of N-P-N (NPN).

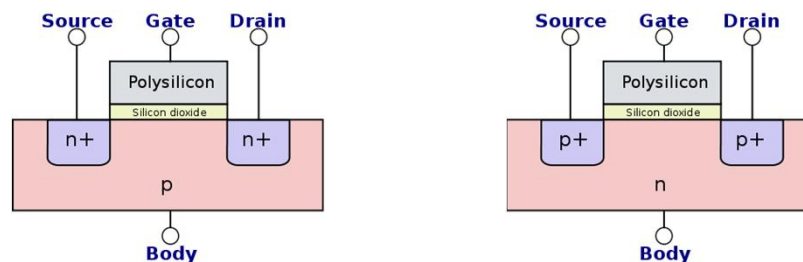


39

WHAT IS A SEMICONDUCTOR

■ Transistor

- For MOSFET, they are NMOS and PMOS



NMOS, n-type semiconductor

PMOS, n-type semiconductor

40

WHAT IS A SEMICONDUCTOR

- Vacuum tubes ruled in first half of 20th century Large, expensive, power-hungry, unreliable

Transistor Advantages

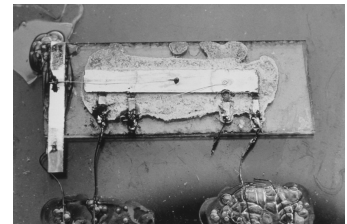
- Highly automated manufacturing processes, resulting in low per-unit cost.
- Extremely long life.
- Higher reliability and greater physical ruggedness.
- Small size and minimal weight, allowing the development of miniaturized electronic devices.
- Lower possible operating voltages, making transistors suitable for small, battery-powered applications.

41

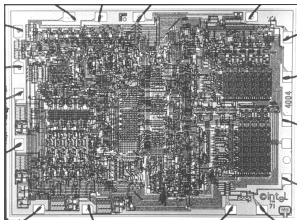
A BRIEF HISTORY

Semiconductor History

- 1958, Jack Kilby at Texas Instruments and Robert Noyce at Fairchild Camera, came up with a solution to the problem of large numbers of components, and the integrated circuit was developed.
- 1970's processes usually had only nMOS transistors → Inexpensive, but consume power while idle
- 1970, The 1st microprocessor was invented at Intel, first commercial in 1975.

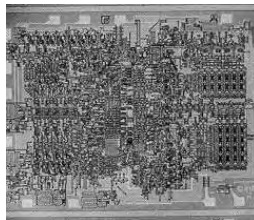


The 1st IC (Flip-flop using two transistors), © Texas Instruments

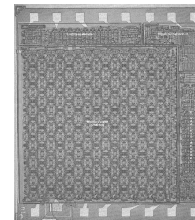


Intel 4004, CPU 4-bit

[© Intel Museum]



Intel 8080, CPU 8-bit



Intel 1101 256-bit SRAM

[Vadasz69]
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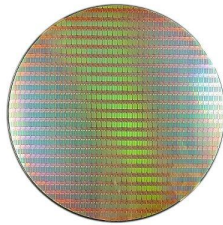
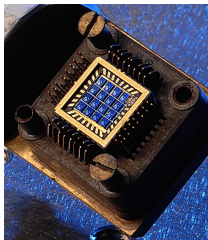
42

A BRIEF HISTORY

■ Semiconductor History

- Instead of making transistors one-by-one, several transistors could be made at the same time, on the same piece of semiconductor (a silicon wafer).
- Not only transistors, but other electric components such as resistors, capacitors and diodes could be made by the same process with the same materials.

→ IC Fabrication → Semiconductor Fabrication (discussed later on)



43

A BRIEF HISTORY

■ The 1st PC



- The IBM PC model 5150 was announced at a press conference in New York on August 12, 1981 and became available for purchase in early Fall 1981.
- Specification ICM PC 5150:
 - Cost for base model: \$2880.
 - This included 64 kilobytes of RAM and a single-sided 160K 5.25" floppy drive.
 - The IBM PC was powered by a 4.77 MHz Intel 8088 processor.

44

A BRIEF HISTORY

- The semiconductor industry has been successful in its consistent efforts to reduce feature size on a chip.

■ 2010

- Intel Core i7 microprocessor
 - 2.3 billion transistors
 - 45nm process
 - Up to 4-core
- 64 Gb Flash memory
 - > 16 billion transistors



[Trinh09]

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■ 2021



45

GROWTH RATE

- 53% compound annual growth rate over 50 years
 - No other technology has grown so fast so long
- Driven by miniaturization of transistors
 - Smaller is cheaper, faster, lower in power!
 - Revolutionary effects on society



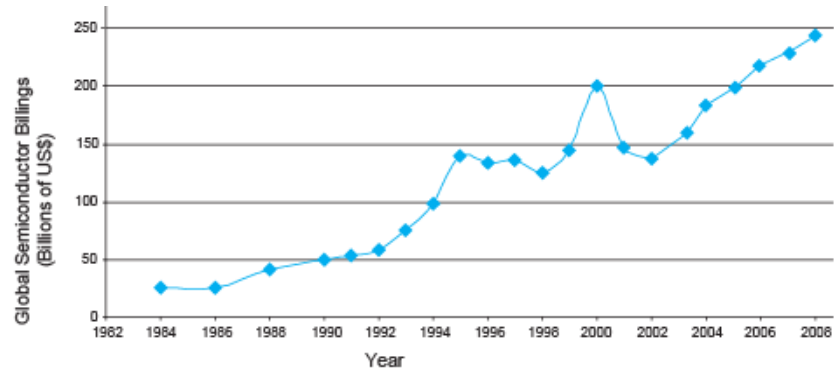
[Moore65]

Electronics Magazine

46

ANNUAL SALES

- $>10^{19}$ transistors manufactured in 2008
 - 1 billion for every human on the planet



47

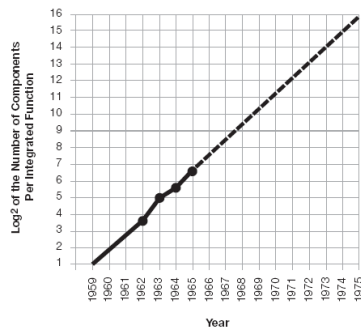
TRANSISTOR TYPES

- Bipolar transistors (BJT)
 - npn or pnp silicon structure
 - Small current into very thin base layer controls large currents between emitter and collector
 - Base currents limit integration density
- Metal Oxide Semiconductor Field Effect Transistors (MOSFET)
 - nMOS and pMOS MOSFETS
 - Voltage applied to insulated gate controls current between source and drain
 - Low power allows very high integration

48

MOORE'S LAW: THEN

- 1965: Gordon Moore plotted transistor on each chip
 - Fit straight line on semilog scale
 - Transistor counts have doubled every 26 months



Integration Levels

SSI: 10 gates

MSI: 1000 gates

LSI: 10,000 gates

VLSI: > 10k gates

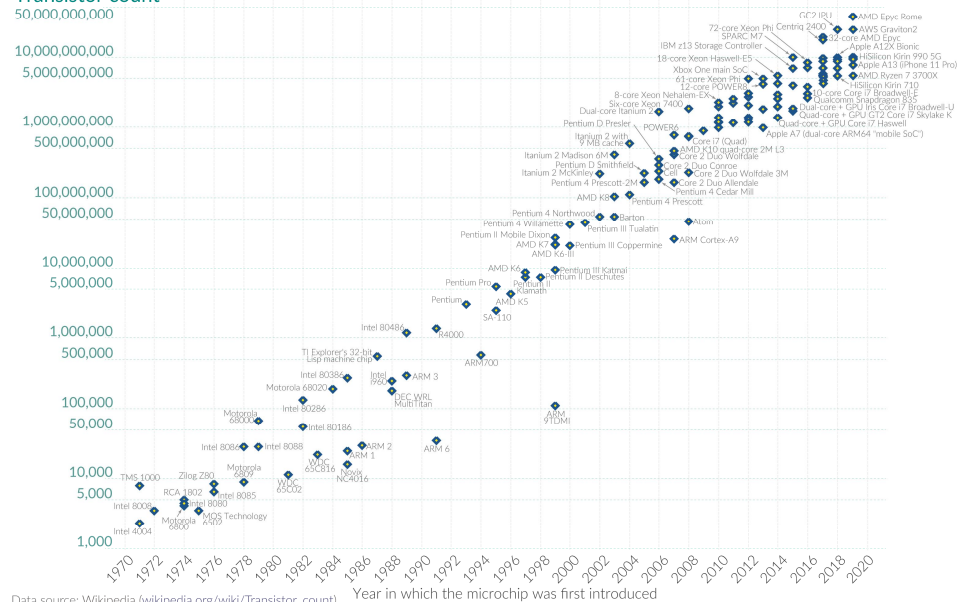
49

Moore's Law: The number of transistors on microchips doubles every two years

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

Our World
in Data

Transistor count



Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)

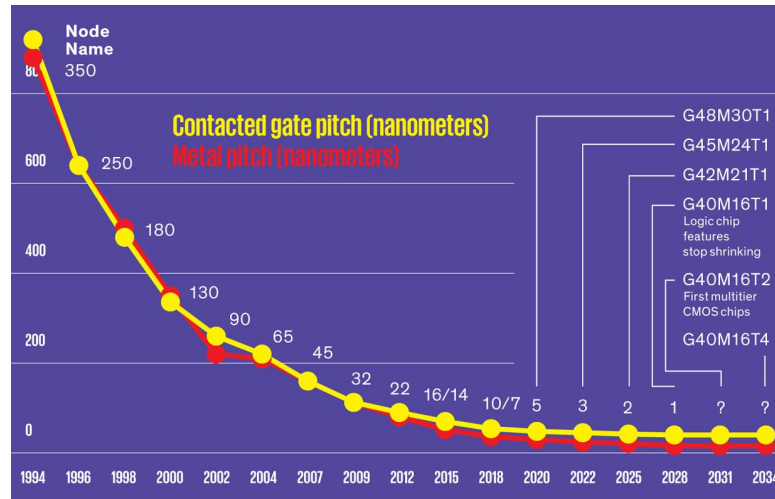
OurWorldinData.org – Research and data to make progress against the world's largest problems.

Licensed under CC-BY by the authors Hannah Ritchie and Max Roser.

50

CMOS TECHNOLOGY

Semiconductor Scaling Trends

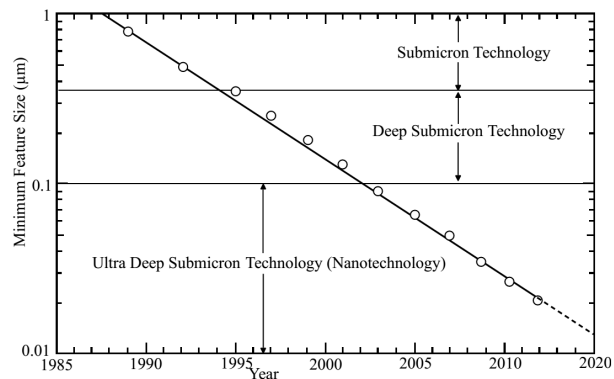


51

CMOS TECHNOLOGY

Categorization of CMOS Technology

- Minimum feature size as a function of time:



- Categories of CMOS technology:

1. Submicron technology – $L_{min} \geq 0.35$ microns
2. Deep Submicron technology (DSM) – $0.1 \text{ microns} \leq L_{min} \leq 0.35 \text{ microns}$
3. Ultra-Deep Submicron technology (UDSM) – $L_{min} \leq 0.1 \text{ microns}$

52

CMOS TECHNOLOGY

Why CMOS Technology?

- Comparison of BJT and MOSFET technology from an analog viewpoint:

Comparison Feature	BJT	MOSFET
Cutoff Frequency (f_T)	100 GHz	50 GHz (0.25 μ m)
Noise (thermal about the same)	Less 1/f	More 1/f
DC range of operation	9 decades of exponential current versus V_{BE}	2-3 decades of square law behavior
Transconductance (same current)	Larger by 10X	Smaller by 10X
Small signal output resistance	Slightly larger	Smaller for short channel
Switch implementation	Poor	Good
Capacitor	Voltage dependent	More options
Performance/Power ratio	High	Low
Technology Improvement	Slower	Faster

53

CMOS TECHNOLOGY

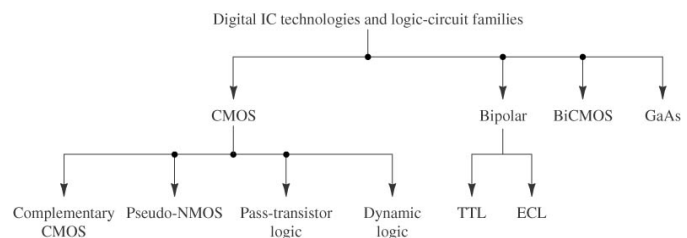
Why CMOS Technology?

Advantages

- Virtually, no DC power consumed
- No DC path between power and ground
- Excellent noise margins ($V_{OL}=0$, $V_{OH}=V_{DD}$)
- Inverter has sharp transfer curve

Drawbacks

- Requires more transistors
- Process is more complicated
- pMOS size larger to achieve electrical symmetry



54

FUNDAMENTAL OF IC PROCESS STEPS

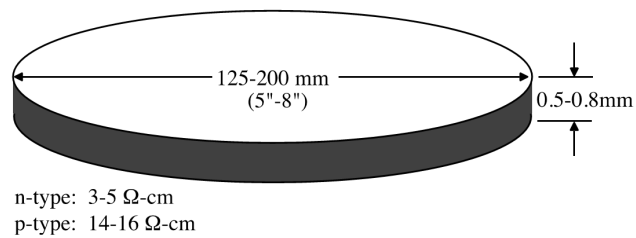
Basic Steps for a CMOS Submicron Process

- Oxide growth
- Thermal diffusion
- Ion implantation
- Deposition
- Etching
- Epitaxy

Photolithography

- Photolithography is the means by which the above steps are applied to selected areas of the silicon wafer.

Silicon Wafer

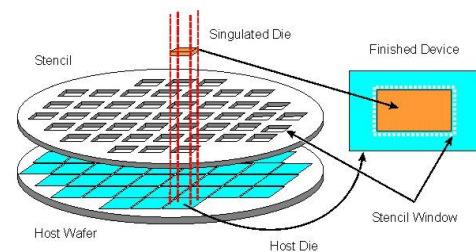
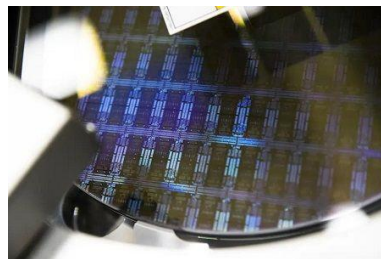
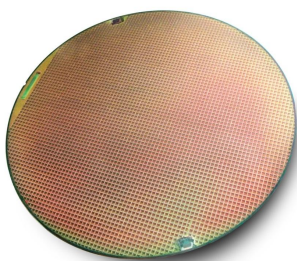


Wafer

55

FUNDAMENTAL OF IC PROCESS STEPS

Wafer

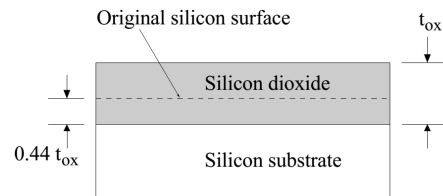


56

FUNDAMENTAL OF IC PROCESS STEPS

Oxidation

- Description: Oxidation is the process by which a layer of silicon dioxide is grown on the surface of a silicon wafer.



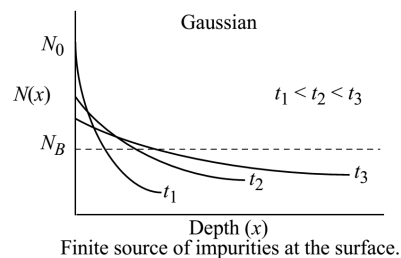
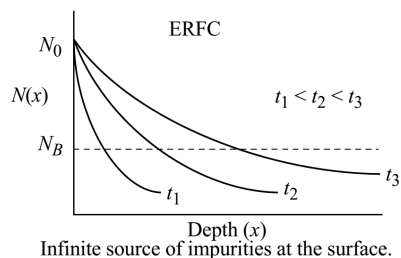
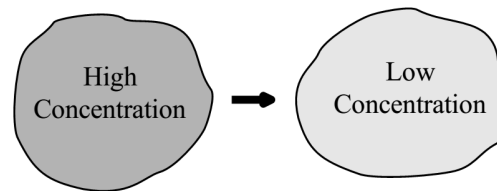
- Uses:
 - Protect the underlying material from contamination
 - Provide isolation between two layers.
- Very thin oxides (100Å to 1000Å) are grown using dry oxidation techniques.
- Thicker oxides (>1000Å) are grown using wet oxidation techniques.

57

FUNDAMENTAL OF IC PROCESS STEPS

Diffusion

- Diffusion is the movement of impurity atoms at the surface of the silicon into the bulk of the silicon.
- Always in the direction from higher concentration to lower concentration.
- Diffusion is typically done at high temperatures: 800 to 1400°C

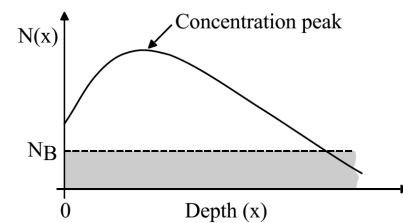
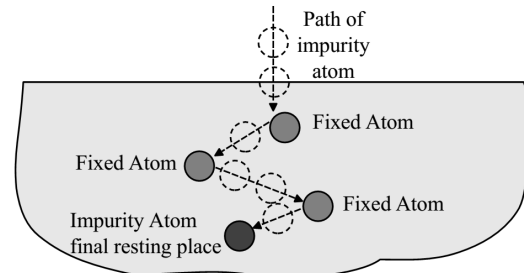


58

FUNDAMENTAL OF IC PROCESS STEPS

Ion implantation

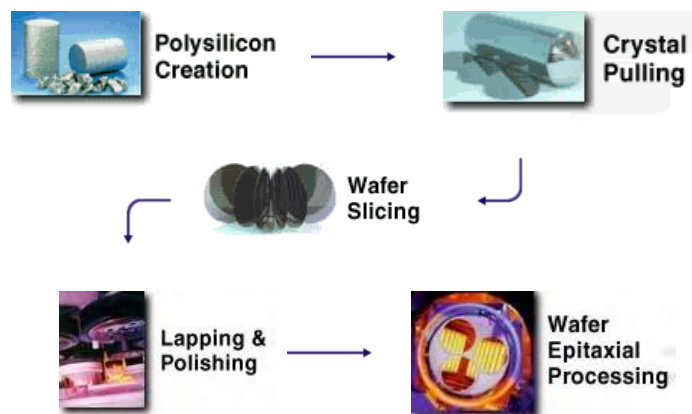
- Ion implantation is the process by which impurity ions are accelerated to a high velocity and physically lodged into the target material.
- Annealing is required to activate the impurity atoms and repair the physical damage to the crystal lattice. This step is done at 500 to 800°C.
- Ion implantation is a lower temperature process compared to diffusion.
- Can implant through surface layers, thus it is useful for field-threshold adjustment.
- Can achieve unique doping profile such as buried concentration peak.
- Generally implant first then use diffusion to achieve the well or active area.



59

FUNDAMENTAL OF IC PROCESS STEPS

Silicon Wafer Production

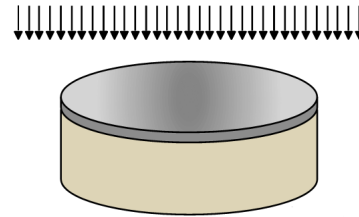


60

FUNDAMENTAL OF IC PROCESS STEPS

Deposition

- Deposition is the means by which various materials are deposited on the silicon wafer.
- Examples:
 - Silicon nitride (Si_3N_4)
 - Silicon dioxide (SiO_2)
 - Aluminum
 - Polysilicon
- There are various ways to deposit a material on a substrate:
 - Chemical-vapor deposition (CVD)
 - Low-pressure chemical-vapor deposition (LPCVD)
 - Plasma-assisted chemical-vapor deposition (PECVD)
 - Sputter deposition
- Material that is being deposited using these techniques covers the entire wafer and requires no mask.



61

FUNDAMENTAL OF IC PROCESS STEPS

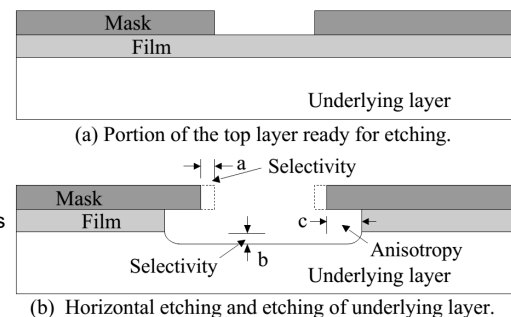
Etching

- Etching is the process of selectively removing a layer of material.
- When etching is performed, the etchant may remove portions or all of:
 - The desired material, The underlying layer, The masking layer
- Important considerations:
 - Anisotropy* of the etch is defined as,

$$A = 1 - (\text{lateral etch rate} / \text{vertical etch rate})$$
 - Selectivity* of the etch (film to mask and film to substrate) is defined as

$$S_{\text{film-mask}} = \text{mask etch rate} / \text{film etch rate}$$

$$A = 1 \text{ and } S_{\text{film-mask}} = \infty \text{ are desired}$$
- There are basically two types of etches:
 - Wet etch which uses chemicals
 - Dry etch which uses chemically active ionized gases.

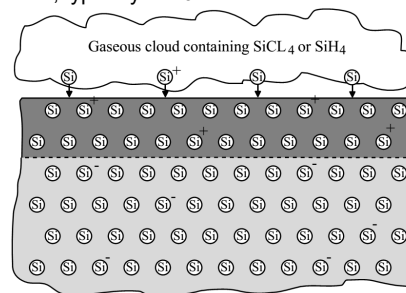


62

FUNDAMENTAL OF IC PROCESS STEPS

Epitaxy

- Epitaxial growth consists of the formation of a layer of single-crystal silicon on the surface of the silicon material so that the crystal structure of the silicon is continuous across the interfaces.
 - It is done externally to the material as opposed to diffusion which is internal
 - The epitaxial layer (epi) can be doped differently, even opposite to the material on which it is grown
 - It is accomplished at high temperatures using a chemical reaction at the surface
 - The epi layer can be any thickness, typically 1 -20 microns



63

FUNDAMENTAL OF IC PROCESS STEPS

Photolithography

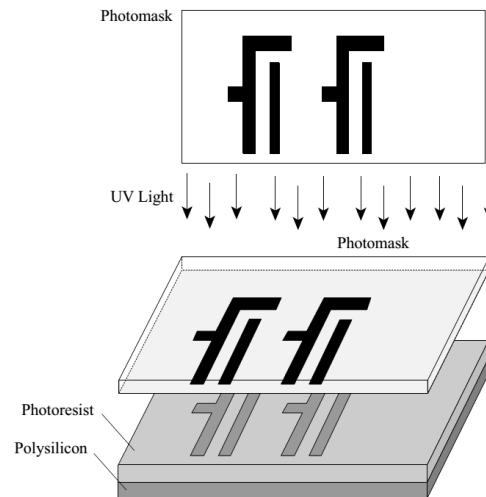
- Components:
 - Photoresist material
 - Mask
 - Material to be patterned (e.g., oxide)
- Positive photoresist: Areas exposed to UV light are soluble in the developer
- Negative photoresist: Areas not exposed to UV light are soluble in the developer
- Steps:
 1. Apply photoresist
 2. Soft bake (drives off solvents in the photoresist)
 3. Expose the photoresist to UV light through a mask
 4. Develop (remove unwanted photoresist using solvents)
 5. Hard bake (≈ 100°C)
 6. Remove photoresist (solvents)

64

FUNDAMENTAL OF IC PROCESS STEPS

Illustration of Photolithography – Exposure

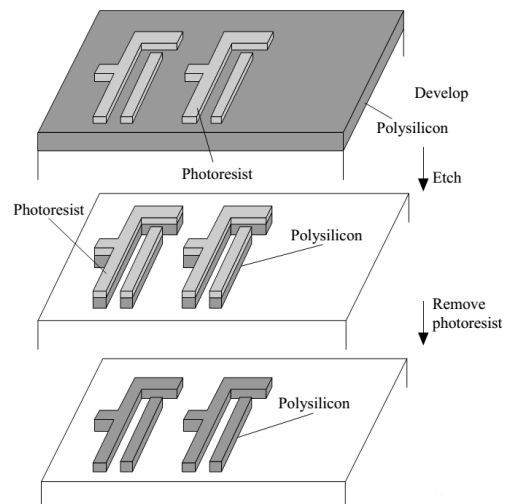
- The process of exposing selective areas to light through a photo-mask is called *printing*.
- Types of printing include:
 - Contact printing
 - Proximity printing
 - Projection printing



65

FUNDAMENTAL OF IC PROCESS STEPS

Illustration of Photolithography – Positive Photoresist

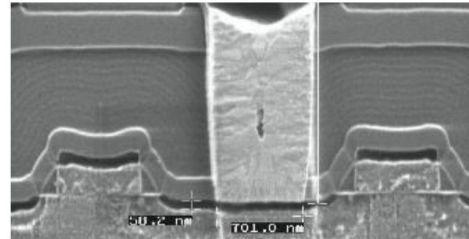


66

TYPICAL SUBMICRON CMOS FABRICATION PROCESS

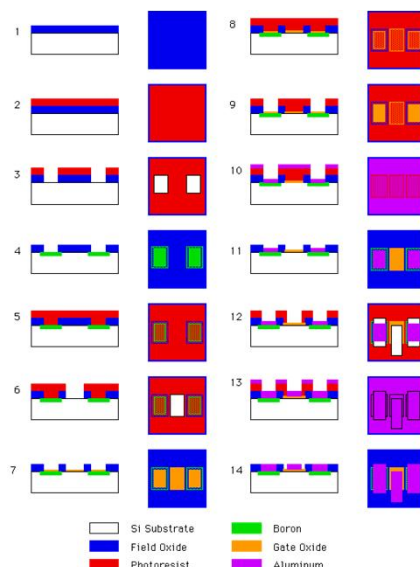
Planarization

- Planarization attempts to minimize the variation in surface height of the wafer.
- Planarization techniques
 - Repeated applications of SOG
 - Resist etch-back – highest areas of oxide are exposed longest to the etchant and therefore erode away the most.
- Influence of planarization on analog design:
 - + Number of levels of metal and the metal integrity depends on planarization
 - + Thin film components at the surface require good planarization
 - + Without planarization, resistance of conductors increases
 - + Planarization at the top level leads to less package induced stress (trimming?)
 - + Planarized passivation helps printing when the depth of field is small.
 - - With planarization, the capacitance of the interdielectric isolation can vary (a good reason to extract capacitance!)
 - - Significant difference in contact aspect ratio (deep versus shallow contacts)



67

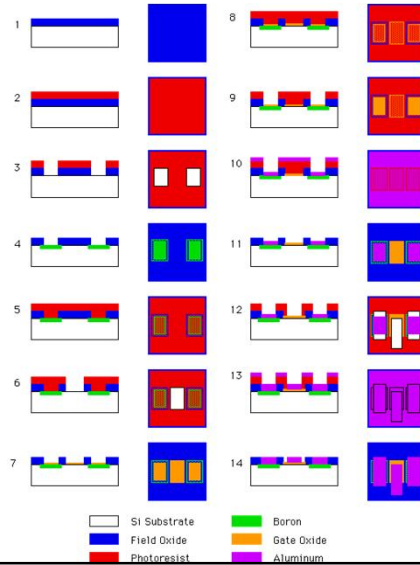
SUMMARY OF FABRICATION PROCESS



1. Clean chip with acetone and isopropanol, and spin on photoresist. Soft-bake the chip in an 85 °C oven for twenty minutes.
2. Expose the photoresist with the boron diffusion mask in UV light for sixty seconds; then develop and rinse in water. Etch the chip in oxide etch for four minutes to remove field oxide in the exposed areas of the chip.
3. Ultrasonically clean the chip in acetone for one minute to remove the photoresist, and rinse in isopropanol. Place the chip in the boron diffusion furnace for thirty minutes at 950 °C. The boron will diffuse into the silicon surfaces, but only a negligible amount will diffuse into the silicon dioxide regions.
4. Etch the chip in boron-glass etch for fifteen seconds, and water for fifteen seconds to remove excess boron. Clean the chip and apply photoresist in the same procedure as above.
5. Expose the photoresist with the gate oxide mask under UV light for sixty seconds, and develop. Etch the exposed field oxide in oxide etch for four minutes and rinse.
6. Remove the photoresist in acetone with the ultrasonic cleaner, and rinse in isopropanol. Place the chip in the gate oxide furnace for thirty minutes at 1000 °C (1 ft³/hr), then switch to nitrogen for ten minutes (1 ft³/hr).
7. Clean the chip and apply photoresist.

68

SUMMARY OF FABRICATION PROCESS



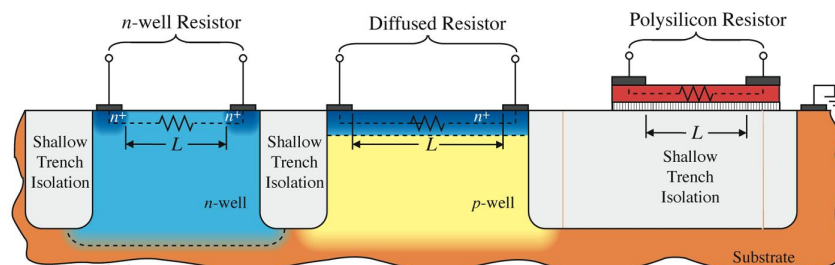
8. Expose the photoresist with the aluminum contact mask under UV light for sixty seconds, and develop.
9. Etch the exposed oxide with oxide etch for four minutes and rinse. Evaporate aluminum onto the entire chip.
10. Remove the photoresist and excess aluminum with the lift off process. Ultrasonically clean the chip in acetone, and the aluminum on the photoresist will dissolve with the photoresist, leaving the aluminum only on the exposed areas.
11. Apply photoresist, expose with the aluminum mask, and develop.
12. Evaporate aluminum on both sides of the chip.
13. Remove photoresist and excess aluminum as in the above process.
14. Complete.

69

CHARACTERISTICS OF A DEEP SUBMICRON CMOS TECHNOLOGY

Resistors

1. Diffused and/or implanted resistors.
2. Well resistors.
3. Polysilicon resistors.
4. Metal resistors.

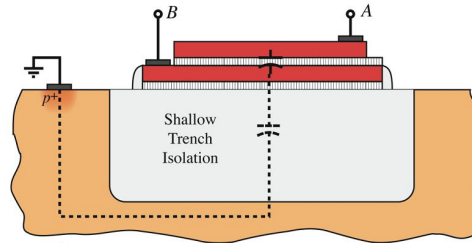


70

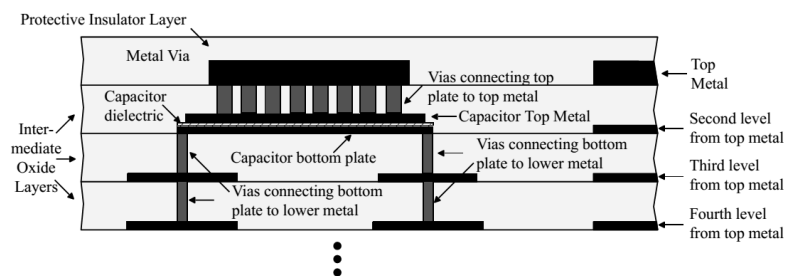
CHARACTERISTICS OF A DEEP SUBMICRON CMOS TECHNOLOGY

Capacitors

- Polysilicon-polysilicon capacitors:



- Metal-metal capacitors:

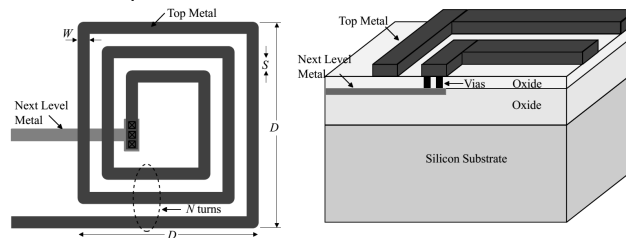


71

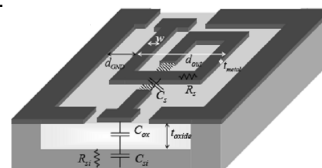
CHARACTERISTICS OF A DEEP SUBMICRON CMOS TECHNOLOGY

Inductors

- Top view and cross-section of a planar inductor:



- Enhanced inductor removing the substrate†:



†M. Raieszadeh, Integrated Inductors on Trenched Silicon Islands, MS Thesis, School of Electrical and Computer Engineering, Georgia Institute of Technology, April 2005

72

TYPICAL DEEP SUBMICRON (DSM) CMOS FABRICATION PROCESS

Major Fabrication Steps for a DSM CMOS Process

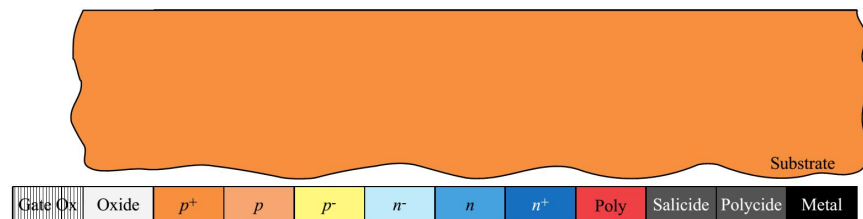
1. p and n wells
2. Shallow trench isolation
3. Threshold shift and anti-punch through implants
4. Thin oxide and gate polysilicon
5. Lightly doped drains and sources
6. Sidewall spacer
7. Heavily doped drains and sources
8. Siliciding (Salicide and Polycide)
9. Bottom metal, tungsten plugs, and oxide
10. Higher level metals, tungsten plugs/vias, and oxide
11. Top level metal, vias and protective oxide

73

TYPICAL DEEP SUBMICRON (DSM) CMOS FABRICATION PROCESS

Starting Material

- The substrate should be highly doped to act like a good conductor.

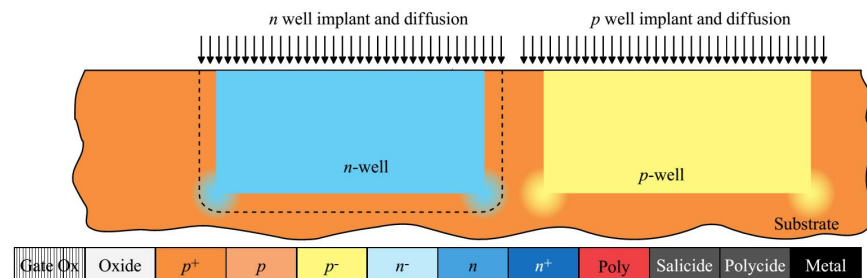


74

TYPICAL DEEP SUBMICRON (DSM) CMOS FABRICATION PROCESS

Step 1 – *n* and *p* wells

- These are the areas where the transistors will be fabricated - NMOS in the *p*-well and PMOS in the *n*-well.
- Done by implantation followed by a deep diffusion.

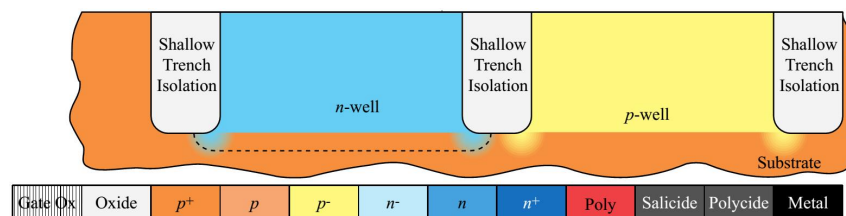


75

TYPICAL DEEP SUBMICRON (DSM) CMOS FABRICATION PROCESS

Step 2 – Shallow Trench Isolation

- The shallow trench isolation (STI) electrically isolates one region/transistor from another.

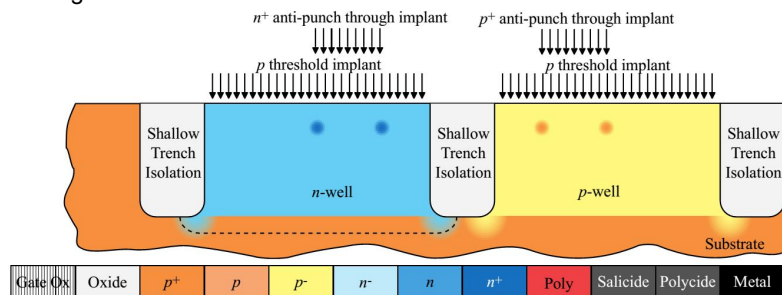
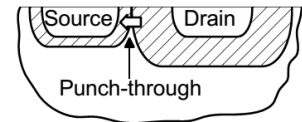


76

TYPICAL DEEP SUBMICRON (DSM) CMOS FABRICATION PROCESS

Step 3 – Threshold Shift and Anti-Punch Through Implants

- The natural thresholds of the NMOS is about 0V and of the PMOS is about – 1.2V. An p-implant is used to make the NMOS harder to invert and the PMOS easier resulting in threshold voltages balanced around zero volts.
- Also an implant can be applied to create a higher-doped region beneath the channels to prevent punch-through from the drain depletion region extending to source depletion region.

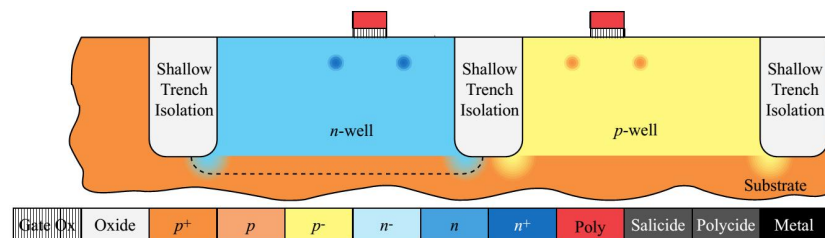


77

TYPICAL DEEP SUBMICRON (DSM) CMOS FABRICATION PROCESS

Step 4 – Thin Oxide and Polysilicon Gates

- A thin oxide is deposited followed by polysilicon. These layers are removed where they are not wanted.

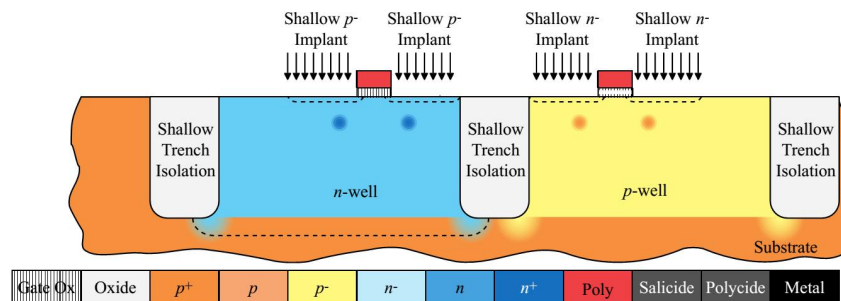


78

TYPICAL DEEP SUBMICRON (DSM) CMOS FABRICATION PROCESS

Step 5 – Lightly Doped Drains and Sources

- A lightly-doped implant is used to create a lightly-doped source and drain next to the channel of the MOSFETs.

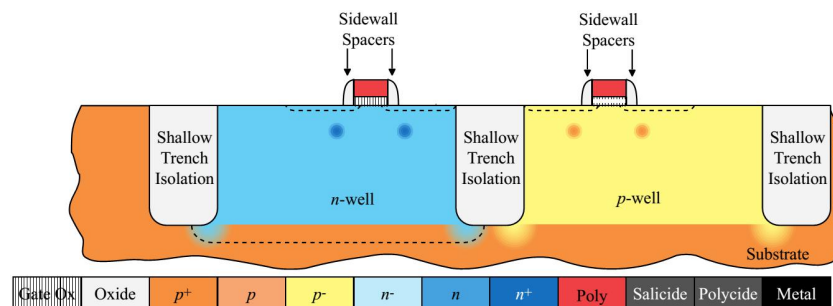


79

TYPICAL DEEP SUBMICRON (DSM) CMOS FABRICATION PROCESS

Step 6 – Sidewall Spacers

- A layer of dielectric is deposited on the surface and removed in such a way as to leave “sidewall spacers” next to the thin-oxide-polysilicon-polycide sandwich. These sidewall spacers will prevent the part of the source and drain next to the channel from becoming heavily doped.

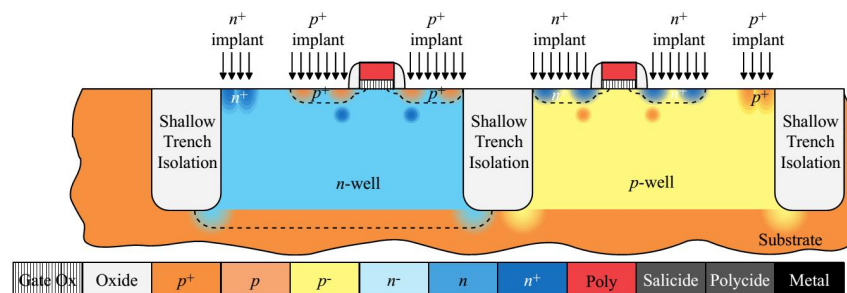


80

TYPICAL DEEP SUBMICRON (DSM) CMOS FABRICATION PROCESS

Step 7 – Implantation of the Heavily Doped Sources and Drains

- Note that not only does this step provide the completed sources and drains but allows for ohmic contact into the wells and substrate.

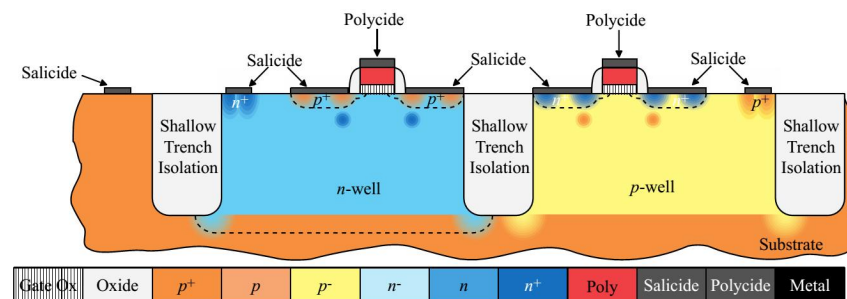


81

TYPICAL DEEP SUBMICRON (DSM) CMOS FABRICATION PROCESS

Step 8 – Siliciding (Salicide and Polycide)

- This step reduces the resistance of the bulk diffusions and polysilicon and forms an ohmic contact (a junction between two conductors that has a linear current-voltage (I-V) curve as with Ohm's law) with material on which it is deposited.
- Salicide = Self-aligned silicide

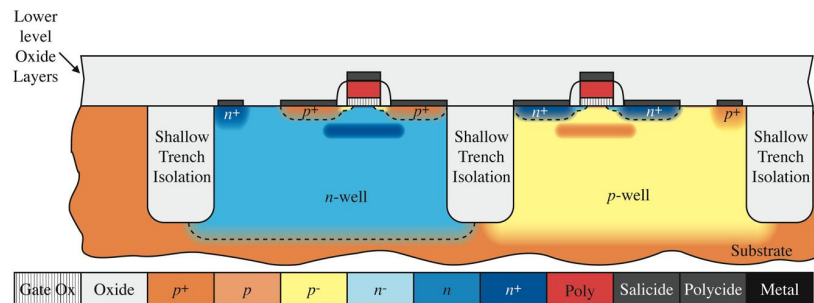


82

TYPICAL DEEP SUBMICRON (DSM) CMOS FABRICATION PROCESS

Step 9 – Intermediate Oxide Layer

- An oxide layer is used to cover the transistors and to planarize the surface.

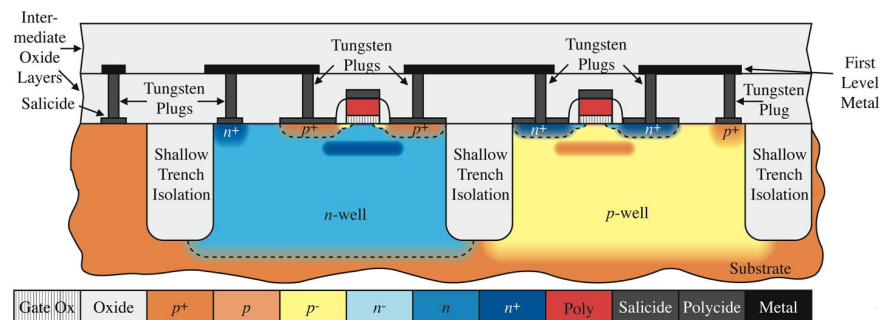


83

TYPICAL DEEP SUBMICRON (DSM) CMOS FABRICATION PROCESS

Step 10- First-Level Metal

- Tungsten plugs are built through the lower intermediate oxide layer to provide contact between the devices, wells and substrate to the first-level metal.

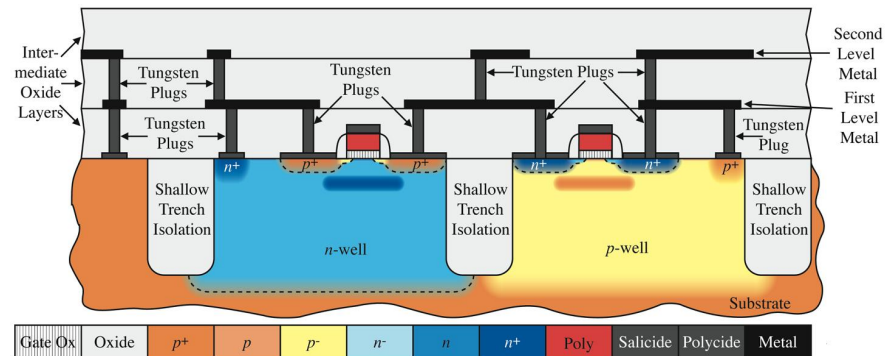


84

TYPICAL DEEP SUBMICRON (DSM) CMOS FABRICATION PROCESS

Step 11 – Second-Level Metal

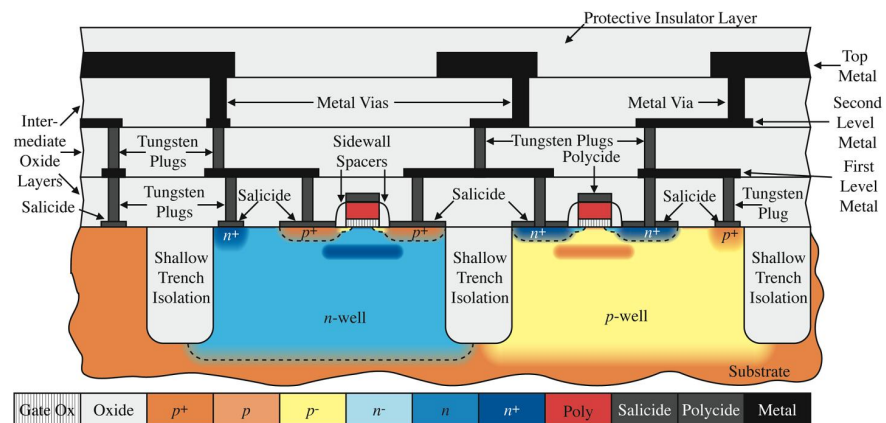
- The previous step is repeated for the second-level metal.



85

TYPICAL DEEP SUBMICRON (DSM) CMOS FABRICATION PROCESS

Completed Fabrication



86

TYPICAL DEEP SUBMICRON (DSM) CMOS FABRICATION PROCESS

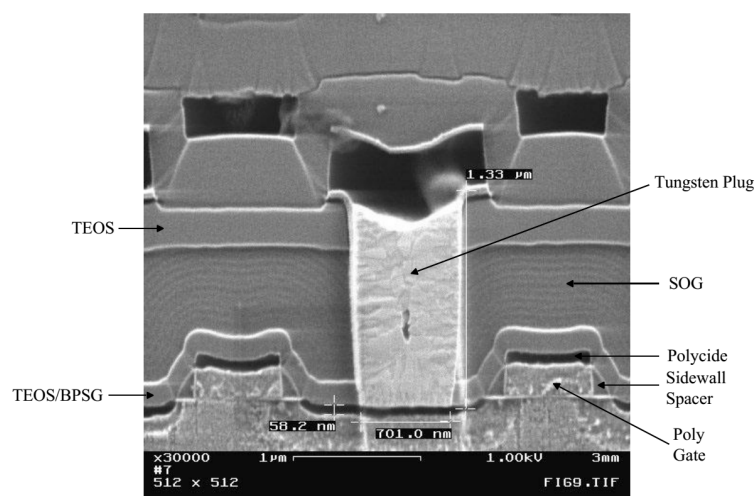
Completed Fabrication

- After multiple levels of metal are applied, the fabrication is completed with a thicker toplevel metal and a protective layer to hermetically seal the circuit from the environment.
- Note that metal is used for the upper level metal vias. The chip is electrically connected by removing the protective layer over large bonding pads.

87

TYPICAL DEEP SUBMICRON (DSM) CMOS FABRICATION PROCESS

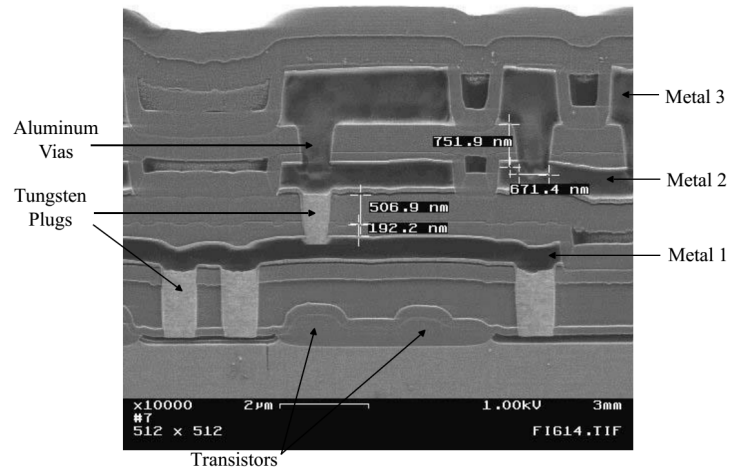
Scanning Electron Microscope (SEM) of a MOSFET Cross-section



88

TYPICAL DEEP SUBMICRON (DSM) CMOS FABRICATION PROCESS

Scanning Electron Microscope Showing Metal Levels and Interconnect



89

ULTRA-DEEP SUBMICRON (UDSM) CMOS TECHNOLOGY

UDSM Technology

- $L_{min} \leq 0.1$ microns
- Minimum feature size less than 100 nanometers
- Today's state of the art:
 - 22 nm drawn length
 - 5 nm lateral diffusion (12 nm gate length)
 - 1 nm transistor gate oxide
 - 8 layers of copper interconnect
- Specialized processing is used to increase drive capability and maintain low off currents

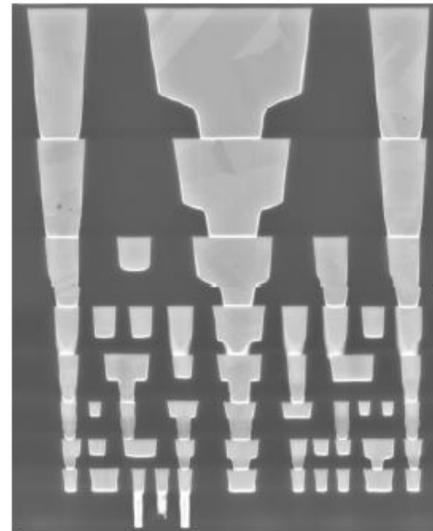
90

ULTRA-DEEP SUBMICRON (UDSM) CMOS TECHNOLOGY

UDSM Metal and Interconnects

Physical aspects:

Layer	Pitch (nm)	Thickness	Aspect Ratio
Isolation	220	230	-
Polysilicon	220	90	-
Contacted Gate Pitch	220	-	-
Metal 1	210	170	1.6
Metal 2	210	190	1.8
Metal 3	220	200	1.8
Metal 4	280	250	1.8
Metal 5	330	300	1.8
Metal 6	480	430	1.8
Metal 7	720	650	1.8
Metal 8	1080	975	1.8



91

ULTRA-DEEP SUBMICRON (UDSM) CMOS TECHNOLOGY

What are the Advantages of UDSM CMOS Technology?

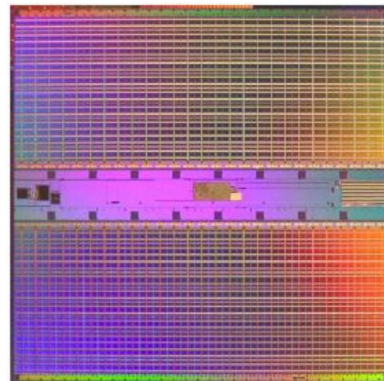
Digital Viewpoint:

- Improved I_{on}/I_{off}
- Reduced gate capacitance
- Higher drive current capability
- Reduced interconnect density
- Reduction of active power

Analog Viewpoint:

- More levels of metal
- Higher f_T
- Higher capacitance density
- Reduced junction capacitance per g_m
- More speed

70 Mbit SRAM chip

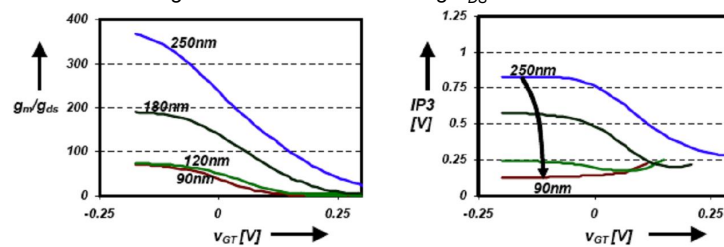


92

ULTRA-DEEP SUBMICRON (UDSM) CMOS TECHNOLOGY

What are the Disadvantages of UDSM CMOS Technology

- Reduction in power supply resulting in reduced headroom (refer headroom in slide P3)
- Gate leakage currents
- Reduced small-signal intrinsic gains
- Increased nonlinearity
- Increased noise and power matching (smaller area)
- Intrinsic gain as a function of the gate overdrive for decreasing $V_{DS}^{\dagger}$:



[†] Anne-Johan Annema, et. Al., "Analog Circuits in Ultra-Deep-Submicron CMOS," *IEEE J. of Solid-State Circuits*, Vol. 40, No. 1, Jan. 2005, pp. 132-143.

93

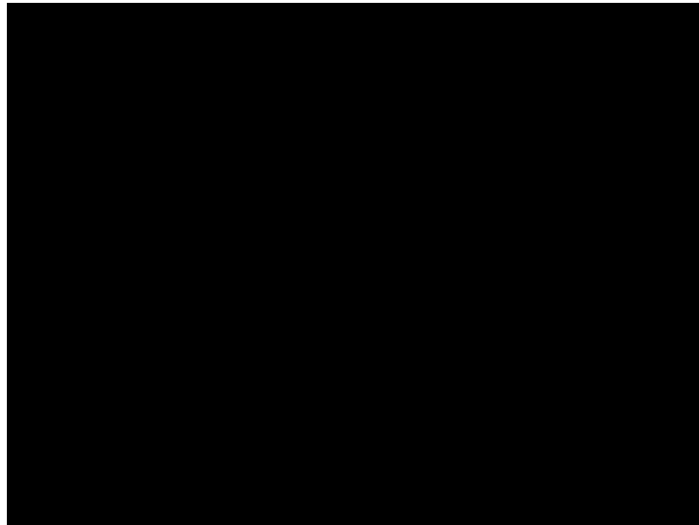
ULTRA-DEEP SUBMICRON (UDSM) CMOS TECHNOLOGY

UDSM CMOS Technology Summary

- Increased transconductance and frequency capability
- Low power supply voltages
- Reduced parasitics
- Gate leakage causes challenges for analog applications of UDSM technology
 - Can no longer use the MOSFET for capacitance
 - Conflict between matching and gate leakage
- Other issues
 - Noise
 - Zero temperature coefficient behavior
 - Etc.

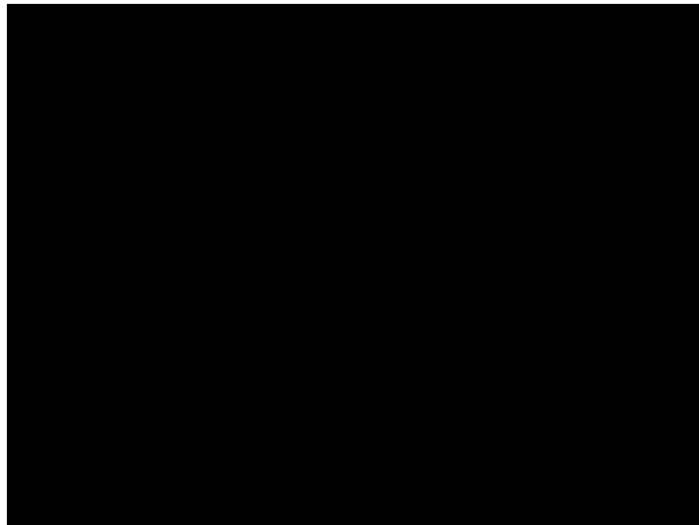
94

VIDEO ON CMOS TECHNOLOGY AND FABRICATION



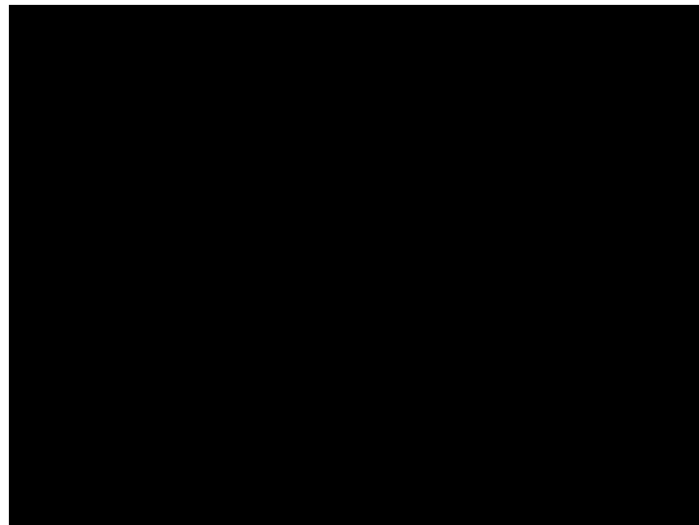
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VIDEO ON CMOS TECHNOLOGY AND FABRICATION



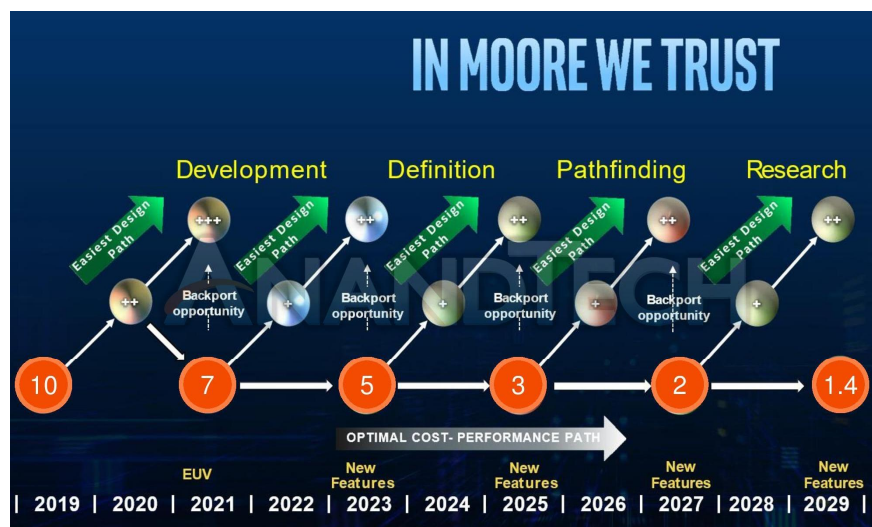
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VIDEO ON CMOS TECHNOLOGY AND FABRICATION



97

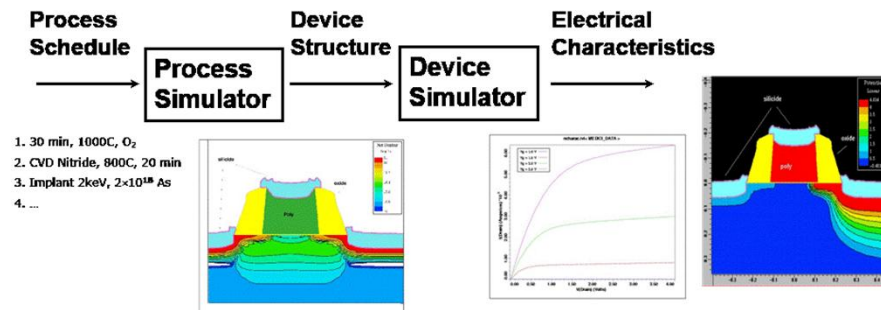
FUTURE CMOS TECHNOLOGY



98

SEMICONDUCTOR PROCESS AND DEVICE SIMULATION

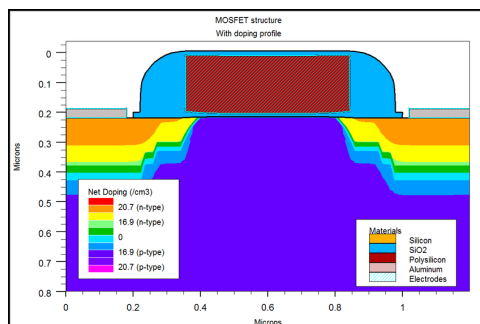
■ TCAD (Technology CAD)



99

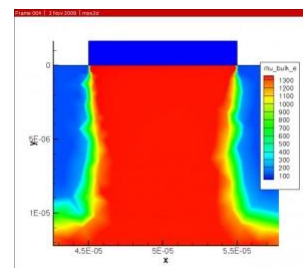
SEMICONDUCTOR PROCESS AND DEVICE SIMULATION

■ TCAD (Silvaco, Synopsys): licensed, high cost



Web: <https://silvaco.com/>

DEVSIM: free and open-source



Web: <https://devsim.org/>

100



**THE END
(THIS PART)**

101